

PAST EXAMINATION QUESTIONS (FP2)

TOPIC 11: MACLAURIN'S SERIES

1. (a) Starting from the definition of $\tanh$ in terms of exponentials, prove that $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$. [3]
- (b) Given that $y = \tanh^{-1} \left(\frac{1-x}{2+x} \right)$, show that $(2x+1) \frac{dy}{dx} + 1 = 0$. [4]
- (c) Hence find the first three terms in the Maclaurin's series for $\tanh^{-1} \left(\frac{1-x}{2+x} \right)$ in the form $a \ln 3 + bx + cx^2$, where a, b and c are constants to be determined. ($a = \frac{1}{2}, b = -1, c = 1$) [5]

Specimen Paper 2020/2/7

2. Given that $y = e^x \cos x$, prove that $\frac{d^2 y}{dx^2} = -2e^x \sin x$, and hence find $\frac{d^3 y}{dx^3}$. [3]

Obtain Maclaurin's series for $e^x \cos x$ up to and including the terms in x^3 . [2]

Nov 1998/1/4

3. [In this question, you may use the result $\frac{d}{dx}(\sec x) = \sec x \tan x$.]

It is given that $y = \ln(\sec x)$.

- (i) Prove that $\frac{d^2 y}{dx^2} = \sec^2 x$. [1]
- (ii) Find Maclaurin's series for y in ascending powers of x , up to and including the term in x^4 . ($\frac{1}{2}x^2 + \frac{1}{12}x^4$) [4]

Nov 1997/1/12

4. (b) Given that $y = \tan \left(3x + \frac{1}{4}\pi \right)$, show that
- (i) $\frac{dy}{dx} = 3(1 + y^2)$. [1]
- (ii) $\frac{d^3 y}{dx^3} = 6y \left(\frac{d^2 y}{dx^2} \right) + 6 \left(\frac{dy}{dx} \right)^2$ [2]
- Find Maclaurin's series for $\tan \left(3x + \frac{\pi}{4} \right)$ up to including the term in x^3 . ($1 + 6x + 18x^2 + 72x^3 + \dots$) [4]

Nov 1996/1/19

5. It is given that $y = \frac{1}{1+\sin 2x}$. Show that, when $x = 0$, $\frac{d^2 y}{dx^2} = 8$. [5]

Find the first three terms in Maclaurin's series for y . [2]

- (i) Use the series to obtain an approximate value for $\int_{-0.1}^{0.1} y dx$, giving your answer correct to four decimal places. (0.2027) [3]

- (ii) Find the first two terms of Maclaurin's series for $\frac{dy}{dx}$. ($-2 + 8x$) [2]

Nov 1995/1/19

Exercise 7

6. Let $y = e^{-2x} \sin 3x$.

(i) Show that $\frac{dy}{dx} = 3e^{-2x} \cos 3x - 2y$, and hence show that

$$\frac{d^2y}{dx^2} = -13y - 4 \frac{dy}{dx}. \quad [5]$$

(ii) Find Maclaurin's series for y , up to including the terms in x^4 . ($y = 3x - 6x^2 + \frac{3}{2}x^3 + 5x^4 + \dots$) [7]

June 1995/1/19

7. A curve C is given by the equation $y^3 + y^2 + y = x^2 - 2x$.

(i) Show that the point $(3,1)$ is the only point of intersection of the line $x = 3$ and the curve. [3]

(ii) Show that the tangent to C at the point $(-1,1)$ has equation $2x + 3y - 1 = 0$. [4]

(iii) Show that, at the origin, $\frac{dy}{dx} = -2$ and $\frac{d^2y}{dx^2} = -6$, and give Maclaurin's series for y as far as the term in x^2 . [5]

Nov 1994/1/19

8. It is given that, at any point on the graph of $y = f(x)$, $\frac{dy}{dx} = \sqrt{1 + y^3}$.

(i) Show that $\frac{d^2y}{dx^2} = \frac{3}{2}y^2$. [3]

(ii) Find expressions for $\frac{d^3y}{dx^3}$ and $\frac{d^4y}{dx^4}$ in terms of y and $\frac{dy}{dx}$. ($3y \frac{dy}{dx}; \frac{9}{2}y^3 + 3 \left(\frac{dy}{dx}\right)^2$) [2]

(iii) The graph of $f(x)$ passes through the origin. Show that, providing x is sufficiently small for powers of x higher than x^4 to be neglected,

$$y = x + \frac{1}{8}x^4. \quad [3]$$

June 1994/1/18

9. Maclaurin's series for $f(x)$ is

$$1 + 3x + 5x^2 + 7x^3 + \dots$$

Find the values of $f'(0)$ and $f'''(0)$. (3;42) [3]

Nov 1993/1/11

10. Given that $y = \sqrt{1 + 8e^x}$, show that $y \frac{dy}{dx} = 4e^x$. [3]

By further differentiation of this result, or otherwise, show that Maclaurin's series for y in ascending powers of x is

$$3 + \frac{4}{3}x + \frac{10}{27}x^2 + \frac{14}{243}x^3 + \dots \quad [5]$$

~~Expand $\left(3 + \frac{4}{3}x + \frac{10}{27}x^2 + \frac{14}{243}x^3\right)^2$ in powers of x up to and including x^2 , and simplify your answer. Explain briefly how the result can be used as a check on the correctness of the first~~ [4]

~~four terms in the series for y . $(9 + 8x + 4x^2 + \frac{4}{3}x^3)$~~

June 1993/1/18

11. The variables x and y are related by

$$\frac{dy}{dx} = 2xy - 1,$$

and $y = 1$ and $x = 0$.

- (i) Show that, at $x = 0$,

$$\frac{d^3y}{dx^3} = -4 \quad \text{and} \quad \frac{d^4y}{dx^4} = 12. \quad [3]$$

Find Maclaurin's series for y up to and including the term in x^4 , ~~and hence find an approximation to the value of y at $x = 0.1$, giving your answer to an appropriate accuracy.~~ [4]
 ~~$(y = 1 - x + x^2 - \frac{2}{3}x^3 + \frac{1}{2}x^4 + \dots; 0.909)$~~

Nov 1992/1/19

12. Let $y = \sec x$. Given that $\frac{dy}{dx} = \sec x \tan x$, express $\frac{d^2y}{dx^2}$ in terms of $\sec x$ and $\tan x$. [2]
 ~~$(\sec^3 x + \tan^2 x \sec x)$~~

Hence show that, up to the term in x^2 , Maclaurin's series for $\sec x$ is $1 + \frac{1}{2}x^2$. [3]

June 1992/1/9

13. (i) Show that the Maclaurin series for $\ln(\sec x)$, up to and including the term in x^4 , is [6]
 $\frac{1}{2}x^2 + \frac{1}{12}x^4$.
~~(ii) Using the Maclaurin's series in part (i), obtain an approximate value for~~

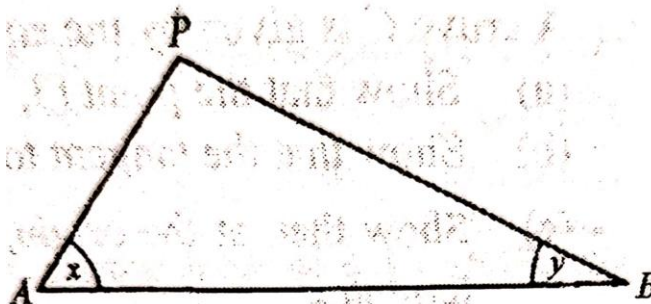
$$\int_0^{0.4} \ln(\sec x) dx,$$

~~Giving four decimal places in your answer. (0.0108)~~

[2]

Nov 1991/1/18 (part)

14. The points A and B are fixed in a plane, and P is a variable point which moves in the plane so that $AP = kBP$, where k is a constant such that $0 < k < 1$. The angles PAB and PBA , measured in radians, are x and y respectively (see diagram).



- (i) Show that $\sin y = k \sin x$. Deduce that $y \leq \sin^{-1}(k)$ for all positions of P .
- (ii) By differentiation of the equation in part (i), or otherwise, show that $\cos y \frac{d^2y}{dx^2} - \sin y \left(\frac{dy}{dx}\right)^2 = -k \sin x$.
- (iii) By using Maclaurin's series, or otherwise, show that, if x is sufficiently small for powers of x above x^3 to be neglected, then $y \approx kx + \frac{k(k^2-1)}{6}x^3$.

June 1991/1/19

15. Let $f(x) = e^{-x} \sin x$. Show that

$$f''(x) = -2\{f'(x) + f(x)\}.$$

$f'''(0)$



[4]

By further differentiation of this result, or otherwise, find the value of $f^{(5)}(0)$, and write down the series expansion of $f(x)$ in ascending powers of x up to and including the term in x^5 . $(-4; x - x^2 + \frac{1}{3}x^3 - \frac{1}{30}x^5)$

[5]

~~Use this series expansion to estimate the value of $\int_0^{\frac{1}{2}} e^{-x} \sin x \, dx$, giving 3 significant figures in your answer. (0.0885)~~

[3]

June 1990/1/17

16. It is given that $\frac{dy}{dx} = y + 2x$ and that $y = 1$ when $x = 0$. Show that $\frac{d^2y}{dx^2} = 3$ when $x = 0$. Find the first three terms in the Maclaurin series for y . $(1 + x + \frac{3}{2}x^2)$

[5]

Nov 1989/1/12

17. Given that $y = \tan\left(\frac{1}{4} + \pi\right)$, show that $\frac{d^2y}{dx^2} = 2y \frac{dy}{dx}$.

[3]

By repeated differentiation of this result, or otherwise, find the series expansion of y in ascending powers of x up to and including the term in x^4 .

[7]

June 1989/1/18 (part)

18. Given that

$$(5x + y) \frac{dy}{dx} = (x + 5y),$$

show that

$$(i) \quad (5x + y)y'' = 1 - (y')^2$$

$$(ii) \quad (5x + y)y''' = -5y'' - 3y'y''$$

[5]

$$\text{where } y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}, y''' = \frac{d^3y}{dx^3}.$$

Given that $y = 2$ when $x = 0$, find the Maclaurin expansion for y up to and including the term in x^3 . $(2 + 5x - 6x^2 + 20x^3)$

[7]

Nov 1988/1/19

19. Given that $y = \ln\left[\frac{1}{2}(1 + e^{-x})\right]$, show that $\frac{dy}{dx} = \frac{1}{2}e^{-y} - 1$.

[4]

By repeated differentiation of this result, or otherwise, show that the series expansion of y in ascending powers of x , up to and including the term in x^4 , is

$$-\frac{1}{2}x + \frac{1}{8}x^2 + kx^4,$$

where the numerical value of k is to be determined. $\left(k = -\frac{1}{192}\right)$

[8]

June 1988/1/19

20. Given that $e^{\cos x}$, show that

$$\frac{d^2y}{dx^2} = -y \cos x - \frac{dy}{dx} \sin x.$$

By further differentiation, or otherwise, show that the first three non-zero terms in the Maclaurin expansion of $e^{\cos x}$ are

$$e - \frac{1}{2}ex^2 + \frac{1}{6}ex^4.$$

Nov 1987/1/20

21. Let $f(x) = \cos^{-1} x$, where $-1 < x < 1$ and $0 < f(x) < \pi$. Show that

$$(1 - x^2)f''(x) - xf'(x) = 0. \quad [4]$$

By differentiating this result three times, or otherwise, show that $f^{(5)}(0) = -9$, and find the first four non-zero terms in the expansion of $f(x)$ in ascending powers of x . $\left(\frac{1}{2}\pi - x - \frac{1}{6}x^3 - \frac{3}{40}x^5\right)$ [8]

June 1987/1/16

22. Given that $y = e^{\tan^{-1} x}$, where $\tan^{-1} x$ denotes the principal value, show that

$$(1 + x^2)\frac{dy}{dx} = y. \quad [2]$$

By repeated differentiation of this result, show also that

$$(1 + x^2)\frac{d^4y}{dx^4} + 6x\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} = \frac{d^3y}{dx^3}. \quad [5]$$

Hence, or otherwise, obtain the expansion of $e^{\tan^{-1} x}$ in ascending powers of x up to and including the term in x^4 , and show that, when x is small enough for powers above the fourth to be neglected.

$$e^{\tan^{-1} x} = e^x - \frac{1}{3}(x^3 + x^4). \quad [5]$$

$$\left(1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{7}{24}x^4\right)$$

June 1986/1/17

23. Let $y = \sqrt{4 + \sin^2 x}$.

(i) Show that $y \frac{dy}{dx} = \frac{1}{2} \sin 2x$. [4]

(ii) By successive differentiation of this result show that

$$y \left(\frac{d^4 y}{dx^4} \right) + 4 \frac{dy}{dx} \frac{d^3 y}{dx^3} + 3 \left(\frac{d^2 y}{dx^2} \right)^2 = -4 \cos 2x. \quad [5]$$

(iii) Find the series expansion of y in ascending powers of x up to and including the term in x^4 . $\left(2 + \frac{1}{4}x^2 - \frac{19}{192}x^4 + \dots\right)$ [4]

Nov 1985/2/21

24. Given that $y = \ln(1 + \sin x)$, find $\frac{dy}{dx}$ in terms of x , and show that

$$\cos x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 0. \quad [4]$$

By further differentiation of this result, or otherwise, show that

$$\cos x \frac{d^4 y}{dx^4} + (1 - 2 \sin x) \frac{d^3 y}{dx^3} - \cos x \frac{d^2 y}{dx^2} = 0. \quad [4]$$

Find the series expansion of $\ln(1 + \sin x)$, in ascending powers of x up to and including the term in x^4 . $\left(\frac{\cos x}{1 + \sin x}; x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{12}x^4 + \dots\right)$ [5]

June 1985/1/21

25. (a) Given that $|x| < 1$, prove that

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right).$$

Hence, or otherwise, find the first three non-zero terms in the series expansion of $\tanh^{-1} x$ in ascending powers of x . $\left(\tanh^{-1} x = x + \frac{1}{3}x^3 + \frac{1}{5}x^5\right)$ [6]

June 1985/2/16 (part)

26. The variables x and y are related by the equation

$$y = \sin^{-1} x - x, \text{ where } 0 \leq x < 1 \text{ and } 0 \leq \sin^{-1} x < \frac{1}{2}\pi.$$

(i) Prove that $\frac{d^2 y}{dx^2} = x \left(1 + \frac{dy}{dx}\right)^3$. [4]

(ii) Find the values of $y, \frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$ at $x = 0$. **(0,0,0)** [2]

(iii) By further differentiation, or otherwise, find the series expansion of y in ascending powers of x up to and including the term in x^3 . $\left(y = \frac{1}{6}x^3 + \frac{3}{40}x^5\right)$ [7]

Nov 1984/1/20

Exercise 7

27. Given that $y = e^{\sqrt{1+x}}$, show that

$$4(1+x) \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = y. \quad [5]$$

By further differentiation of this result, or otherwise, show that the series expansion for y in ascending powers of x up to the term in x^4 is

$$y = e \left(1 + \frac{1}{2}x + \frac{1}{48}x^2 + kx^4 \right),$$

where k is a number whose value is to be found. $\left(k = -\frac{5}{384}\right)$ [8]

June 1984/1/21

28. Given that $y = e^{-\sin 2x}$, prove that

$$\frac{d^2y}{dx^2} + (2 \cos 2x) \frac{dy}{dx} - (4 \sin 2x)y = 0,$$

and obtain the values of y , $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $x = 0$. $(1, -2, 4)$

By further differentiation obtain the values of $\frac{d^2y}{dx^2}$ and $\frac{d^4y}{dx^4}$ when $x = 0$. $(0, -48)$

Hence, or otherwise, obtain the expansion of y in ascending powers of x up to and including the term in x^4 . $(1 - 2x + 2x^2 - 2x^4)$

Nov 1983/2/20

29. Let $y = \tan^{-1}(1+x)$, where the principal value is taken.

(i) Find $\frac{dy}{dx}$ in terms of x , and deduce that

$$(2 + 2x + x^3) \frac{d^2y}{dx^2} + (2 + 2x) \frac{dy}{dx} = 0.$$

$$\left(\frac{dy}{dx} = \frac{1}{1 + (1+x)^2} \right)$$

(ii) By further differentiation of the result in (i), or otherwise, show that

$$(2 + 2x + x^3) \frac{d^5y}{dx^5} + (8 + 8x) \frac{d^4y}{dx^4} + 12 \frac{d^2y}{dx^2} = 0.$$

(iii) Find the series expansion of y in ascending powers of x up to and including the term in x^4 . $\left(y \approx \frac{\pi}{4} + \frac{1}{2}x - \frac{1}{4}x^2 + \frac{1}{12}x^3 - \frac{1}{40}x^5\right)$

June 1983/2/21

30. Show that, if x is so small that x^4 and higher powers of x may be neglected,

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5.$$

Deduce that, for such values of x ,

$$\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{1}{3}x^3 + \frac{1}{5}x^5\right).$$

By giving x a suitable value in this last result, prove that, for large N ,

$$\ln\left(\frac{N+1}{N-1}\right) \approx \frac{2}{N} + \frac{2}{3N^3} + \frac{2}{5N^5}.$$

Hence show that, for large N ,

$$\left(\frac{N+1}{N-1}\right)^N \approx e^2.$$

June 1982/1/21

31. Given that $y = \sqrt{\cos x}$, prove that

$$y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + \frac{1}{2}y^2 = 0.$$

By repeated differentiation of this result, or otherwise, express y as a series of ascending powers of x up to and including the terms in x^4 . $\left(y = 1 - \frac{x^2}{4} - \frac{x^4}{96}\right)$

Nov 1981/2/21

32. Let $y = e^x \sin x$. Show that $\frac{dy}{dx}$ may be expressed as $ce^x \sin(x + \alpha)$, where $c > 0$ and $-\frac{1}{2}\pi < \alpha < \frac{1}{2}\pi$, and state the numerical value of c and α . $\left(c = \sqrt{2}, \alpha = \frac{1}{4}\pi\right)$

Hence, or otherwise, find $\frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \frac{d^4y}{dx^4}$ and $\frac{d^5y}{dx^5}$. $\left(2e^x \cos x; 2\sqrt{2}e^x \sin\left(x + \frac{3}{4}\pi\right); -4e^x \sin x - 4\sqrt{2}e^x \sin\left(x + \frac{1}{4}\pi\right)\right)$

Using these results, or otherwise, expand y as a series of ascending powers of x up to and including the terms in x^5 . $\left(y \approx x + x^2 + \frac{1}{3}x^3 - \frac{1}{30}x^5\right)$

June 1981/1/21

33. Given that $y = e^{\sin^{-1} x}$, prove that

Exercise 7

$$(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - y = 0.$$

Hence, or otherwise, expand $e^{\sin^{-1} x}$ in ascending powers of x up to and including the term in x^3 . $\left(1 + x + \frac{1}{2}x^2 + \frac{1}{3}x^3\right)$

Nov 1980/1/20

34. Given that $y = \sqrt{(\cos x)}$, show that

(i) $4y^2 \left(\frac{dy}{dx}\right)^2 = 1 - y^4,$

(ii) $\frac{d^2y}{dx^2} = -\frac{1}{4}(y^{-3} + y).$

By further differentiation, or otherwise, obtain the expression of y as a series of ascending powers of x up to and including the term in x^4 . $\left(y = 1 - \frac{1}{4}x^2 - \frac{1}{96}x^4\right)$

June 1980/2/21

35. If $y = (\sin^{-1} x)^2$, prove that

$$(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2.$$

By repeated differentiation of this result and use of Maclaurin's theorem, or otherwise, prove that the first three non-zero terms in the expansion of $(\sin^{-1} x)^2$ are

$$x^2 + \frac{1}{3}x^4 + \frac{8}{45}x^6.$$

Nov 1979/2/20

36. Given that $y = \tan^2 x$, express $\frac{dy}{dx}$ in terms of $\tan x$, and hence show that

$$\frac{d^2y}{dx^2} = 2 + 8y + 6y^2.$$

$$\left(\frac{dy}{dx} = 2 \tan x (1 + \tan^2 x)\right)$$

By further differentiation of this result, or otherwise, show that, if x is small enough for powers of x above x^4 to be neglected, then

$$\tan^2 x = x^2 + \frac{2}{3}x^4 + \frac{17}{45}x^6.$$

June 1979/2/22

37. If $f(x) = e^{\cos x}$, prove that

$$f''(x) + \sin x f'(x) + \cos x f(x) = 0.$$

Assuming that $e^{\cos x}$ can be expanded in a series of ascending powers of x , show that, neglecting powers of x above the fourth,

$$e^{\cos x} = e \left(1 - \frac{1}{2}x^2 + \frac{1}{6}x^4\right).$$

Nov 1978/1/21

38. If $y = \ln(\sec x)$, prove that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2 + 1$.

By further differentiation of this result, or otherwise, find the series expansion of y in terms of x up to and including the term in x^4 . $\left(\frac{1}{2}x^2 + \frac{1}{12}x^4\right)$

June 1978/1/21

39. (a) Find $\int \tanh^{-1} x \, dx$. $\left(x \tanh^{-1} x + \frac{1}{2} \ln(1 - x^2) + c\right)$
 (b) Write down the first three non-zero terms in
 (i) the power series expansion of $\frac{\tanh^{-1} x}{x}$, $\left(1 + \frac{1}{3}x^2 + \frac{1}{5}x^4\right)$
 (ii) the binomial expansion of $(1 + \alpha x^2)^\beta$. $\left(1 + \alpha\beta x^2 + \frac{\alpha^2\beta(\beta-1)}{2}x^4\right)$

Find the values of α, β if the answers in (i) and (ii) are identical. $\left(\alpha = -\frac{13}{15}, \beta = -\frac{5}{13}\right)$

Using this identity and the logarithmic equivalent of $\tanh^{-1} \frac{1}{3}$ show that

$$\ln 2 \approx \frac{2}{3} \left(\frac{135}{122}\right)^{\frac{5}{13}}.$$

November 1978/2/17

40. Show that the first five non-zero terms in the expansion of $\ln(1 + x)$ in ascending powers of x are

$$x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5.$$

Express $\tanh y$ in terms of e^y and hence express $\tanh^{-1} x$ in terms of a natural logarithmic function of x . $\left(\frac{e^y - e^{-y}}{e^y + e^{-y}}; \frac{1}{2} \ln \frac{1+x}{1-x}\right)$

Hence, or otherwise, find the first three non-zero terms in the series expansion of $\tanh^{-1} x$. $\left(x + \frac{1}{3}x^3 + \frac{1}{5}x^5\right)$

State, with reasons, whether or not $\cosh^{-1} x$ can be expanded in ascending powers of x . (No)
 MB November 1977/1/20

41. (a) Sketch the graph of f , defined by $f: x \rightarrow \sinh^{-1} x$, where x is real.
 (b) If $y = \sin(\sinh^{-1} x)$, show that

$$(1 + x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0.$$

Hence, or otherwise, find the first three non-zero terms in the expansion of y in ascending powers of x . $\left(y = x - \frac{1}{3}x^3 + \frac{1}{6}x^5\right)$

MB November 1976/2/22

42. Prove that $\sinh^{-1} y = \ln[y + \sqrt{(y^2 + 1)}]$. Deduce that, if y is large and positive, then

$$\sinh^{-1} y \approx \ln(2y) + \frac{1}{4y^2}.$$

MB June 1977/1/20

43. If $y = \ln(1 + \sin x)$, find $\frac{dy}{dx}$ and show that $\frac{d^2y}{dx^2} = -\frac{1}{1+\sin x}$. Find the expansion in ascending powers of x of $\ln(1 + \sin x)$ up to and including the term in x^4 . Deduce that, if powers of x above the fourth are neglected,

$$\ln\left(\frac{1+x}{1+\sin x}\right) = \frac{1}{6}x^3(1-x).$$

June 1977/2/21

44. Find the first three derivatives of $\tan x$ and of $\tan^{-1} x$. Hence obtain, using Maclaurin's theorem, or otherwise, expressions for $\tan x$ and $\tan^{-1} x$ in ascending powers of x as far as the term in x^3 .

$$\left(\sec^2 x, 2 \sec^2 x \tan x, 2 \sec^4 x + 4 \sec^2 x \tan^2 x; \frac{1}{1+x^2}, -\frac{2x}{(1+x^2)^2}, \frac{-2+6x^2}{(1+x^2)^3}; x + \frac{1}{3}x^3, x - \frac{1}{3}x^3\right)$$

June 1976/1/21

45. If $y = \tanh^{-1} x$, prove that $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = 0$.

Differentiate three more times, and find the values of the first five derivatives of $\tanh^{-1} x$ at $x = 0$. **(1, 0, 2, 0, 24)**

Hence, or otherwise, write down the first three non-zero terms in the Maclaurin expansion for $\tanh^{-1} x$. $\left(x + \frac{1}{3}x^3 + \frac{1}{5}x^5\right)$

MB June 1974/1/21

46. Given that $y = e^x \sin x$, express $\frac{dy}{dx}$ in the form $ke^x \sin(x + \alpha)$ stating clearly suitable values for k and α . Express the next two derivatives in similar form. $\left(\sqrt{2}, \frac{1}{4}\pi\right)$

With the aid of Maclaurin's Theorem, or otherwise, express y as a series of ascending powers of x as far as the term in x^3 . $\left(x + x^2 + \frac{1}{3}x^3\right)$

June 1972/2/22 (part)

47. If $y = \tan^{-1} x$, prove that $(1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$.

Differentiate three more times, and find the values of the first five derivatives of $\tan^{-1} x$ at $x = 0$. **(1, 0, -2, 0, 24)**

Hence, or otherwise, find the first three non-zero terms in the Maclaurin expansion of $\tan^{-1} x$. $\left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5\right)$

Nov 1974/2/22

48. The exponential function e^x may be defined as follows:

$$\frac{d}{dx}(e^x) = e^x,$$

$$e^0 = 1.$$

Assuming that e^x can be expressed as an infinite series in ascending powers of x , derive the series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots$$

June 1972/2/21